Topics: 15.6 Triple Integrals; 15.7 Triple Integrals in Cylindrical Coordinates; 15.8 Triple Integrals in Spherical Coordinates

Please put away all electronic devices, including cell phones and calculators.

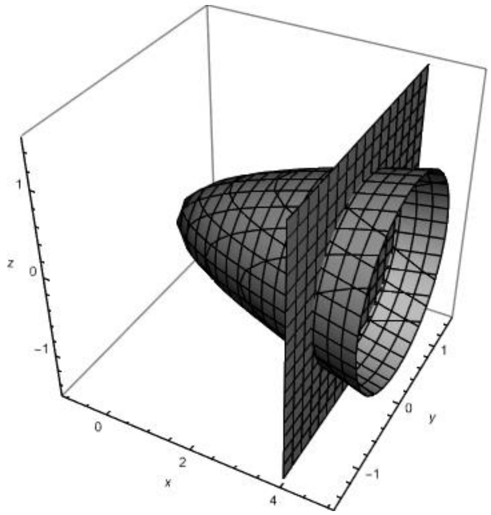
1. Evaluate the triple integral $\iiint_E (xy + z^2)\, dV$ where $E = [0, 2] \times [0, 1] \times [0, 3]$.

$$\begin{aligned}
\int_0^2 \int_0^1 \int_0^3 (xy + z^2) dz dy dx &= \int_0^2 \int_0^1 (3xy + 9) dy dx \\
&= \int_0^2 \left(\frac{3}{2}x + 9\right) dx = 21 \\
\int_0^1 \int_0^2 \int_0^3 (xy + z^2) dz dx dy &= \int_0^1 \int_0^2 (3xy + 9) dx dy \\
&= \int_0^1 (6y + 18) dy = 21
\end{aligned}$$

2. Evaluate $\iiint_E \frac{1}{x^3}\, dV$ where $E = \{(x, y, z) | 0 \le y \le 1, 0 \le z \le y^2, 1 \le x \le z + 1\}$.

$$\begin{aligned}
\iiint_E \frac{1}{x^3}\, dV &= \int_0^1 \int_0^{y^2} \int_1^{z+1} x^{-3} dx dz dy \\
&= \int_0^1 \int_0^{y^2} \frac{1}{2} - \frac{1}{2}(z + 1)^{-2} dz dy \\
&= \int_0^1 \left(\frac{1}{2}y^2 + \frac{1}{2}\frac{1}{1 + y^2} - \frac{1}{2}\right) dy \\
&= \frac{3\pi - 8}{24}
\end{aligned}$$

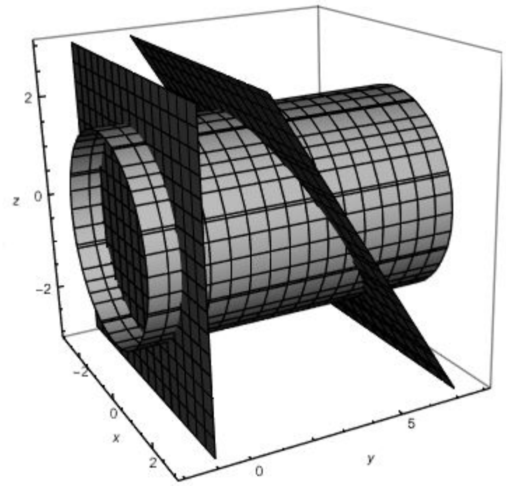
3. Evaluate $\iiint_E x\, dV$ where E is the solid bounded by the paraboloid $x = 4y^2 + 4z^2$ and $x = 4$.



The paraboloid and the plane intersect in the plane $x = 4$ as the circle $4y^2 + 4z^2 = 4$. This solid region projects onto the yz-plane into the disk $y^2 + z^2 \leq 1$. We use cylindrical coordinates (r, θ, x), and the change of variables $y = r\cos\theta, z = r\sin\theta$. Then the region is bounded by $x = 4r^2$ and $x = 4$ which gives the new bounds of integration:

$$\begin{aligned} \iiint_E x\, dV &= \int_0^{2\pi} \int_0^1 \int_{4r^2}^4 xr dx dr d\theta \\ &= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^1 (8r - 8r^5) dr \right) \\ &= \frac{16\pi}{3} \end{aligned}$$

4. Find the volume of the solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.



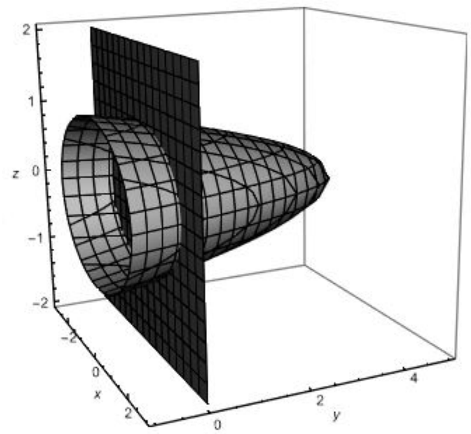
We use cylindrical coordinates (r, θ, y), and the change of variables $x = r\cos\theta$, $z = r\sin\theta$. The solid projects onto the xz-plane as the disk $x^2 + z^2 \leq 4$. Thus, $0 \leq r \leq 2$, and $0 \leq \theta \leq 2\pi$.Then we can write the bounds for y as

$$-1 \leq y \leq 4 - z = 4 - r\sin\theta.$$

Then:

$$\begin{aligned}\iiint_E 1 \; dV &= \int_0^{2\pi} \int_0^2 \int_{-1}^{4-r\sin\theta} r dy dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (5r - r^2 \sin\theta) dr d\theta \\ &= \int_0^{2\pi} (10 - \frac{8}{3}\sin\theta) d\theta = 20\pi.\end{aligned}$$

5. Express the triple integral $\iiint_E f(x, y, z)\, dV$ as an iterated integral in six different ways, where E is the solid bounded by $y = 4 - x^2 - 4z^2$ and $y = 0$.



We have

$$
\begin{aligned}
\iiint_E f(x,y,z)\,dV &= \int_{-1}^{1}\int_{-2\sqrt{1-z^2}}^{2\sqrt{1-z^2}}\int_{0}^{4-x^2-4z^2} f(x,y,z)dydxdz \\
&= \int_{-2}^{2}\int_{-\sqrt{1-(x/2)^2}}^{\sqrt{1-(x/2)^2}}\int_{0}^{4-x^2-4z^2} f(x,y,z)dydzdx \\
&= \int_{-1}^{1}\int_{0}^{4-4z^2}\int_{-\sqrt{4-4z^2-y}}^{\sqrt{4-4z^2-y}} f(x,y,z)dxdydz \\
&= \int_{-2}^{2}\int_{0}^{4-x^2}\int_{-\sqrt{1-x^2/4-y/4}}^{\sqrt{1-x^2/4-y/4}} f(x,y,z)dzdydx \\
&= \int_{0}^{4}\int_{-\frac{1}{2}\sqrt{4-y}}^{\frac{1}{2}\sqrt{4-y}}\int_{-\sqrt{4-4z^2-y}}^{\sqrt{4-4z^2-y}} f(x,y,z)dxdzdy \\
&= \int_{0}^{4}\int_{-\sqrt{4-y}}^{\sqrt{4-y}}\int_{-\sqrt{1-x^2/4-y/4}}^{\sqrt{1-x^2/4-y/4}} f(x,y,z)dzdxdy
\end{aligned}
$$

6. What surface does the equation $z^2 + r^2 = 9$ describe in cylindrical coordinates?

Since $x = r\cos(\theta)$ and $y = r\sin(\theta)$, we have $x^2 + y^2 = r^2\left(\cos^2(\theta) + \sin^2(\theta)\right) = r^2$.Therefore the given equation can be written as

$$
\begin{aligned}
z^2 + r^2 &= 9 \\
z^2 + x^2 + y^2 &= 3^2
\end{aligned}
$$

Hence, the above equation represents a sphere centered at the origin, with radius 3.

7. Evaluate $\iiint_E z\,dV$ where E is enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

The equation of the paraboloid $z = x^2 + y^2$ in cylindrical coordinates is $z = r^2$. We can describe E using cylindrical coordinates.

$$E = \{(r, \theta, z) : 0 \le r \le 2, 0 \le \theta \le 2\pi, r^2 \le z \le 4\}$$

$$\begin{aligned}
\iiint_E z\,dV &= \int_0^{2\pi} \int_0^2 \int_{r^2}^4 z\ r\,dz\,dr\,d\theta \\
&= \int_0^{2\pi} \int_0^2 \frac{z^2}{2} r \Big|_{r^2}^4 dr\,d\theta \\
&= \int_0^{2\pi} \int_0^2 \left(8r - \frac{r^5}{2}\right) dr\,d\theta \\
&= \int_0^{2\pi} (4r^2 - \frac{r^6}{12}\Big|_0^2)\,d\theta \\
&= \int_0^{2\pi} \left[\left(16 - \frac{16}{3}\right) - 0\right] d\theta \\
&= \int_0^{2\pi} \frac{32}{3}\,d\theta \\
&= \frac{32}{3} \cdot 2\pi \\
&= \frac{64\pi}{3}
\end{aligned}$$

8. Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.

We can describe E using cylindrical coordinates.

$$\begin{aligned}
\textbf{Sphere:}\quad r^2 + z^2 &= 2 \\
z &= \pm\sqrt{2 - r^2} \\
z &= +\sqrt{2 - r^2} \quad \text{(Note that we only need the "+" for this problem.)} \\
\textbf{Paraboloid:}\quad z &= r^2
\end{aligned}$$

$$E = \{(r, \theta, z) : 0 \le \theta \le 2\pi, 0 \le r \le 1, r^2 \le z \le \sqrt{2 - r^2}.\}$$

Note that $0 \le r \le 1$ since

$$\begin{aligned}
r^2 + (r^2)^2 &= 2 \\
r^4 + r^2 - 2 &= 0 \\
(r^2 + 2)(r^2 - 1) &= 0 \\
\implies r &= 1
\end{aligned}$$

$$
\begin{aligned}
\text{Volume}(E) &= \iiint_E dV \\
&= \int_0^{2\pi}\int_0^1\int_{r^2}^{\sqrt{2-r^2}} r\,dz\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^1 z\,r\Big|_{r^2}^{\sqrt{2-r^2}} dr\,d\theta \\
&= \int_0^{2\pi}\int_0^1 (r\sqrt{2-r^2}-r^3)\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^1 r\sqrt{2-r^2}\,dr\,d\theta - \int_0^{2\pi}\int_0^1 r^3\,dr\,d\theta
\end{aligned}
$$

To evaluate the first integral in the expression above, we proceed by u-substitution

$$
\begin{cases}
\text{Let } u = 2-r^2 \rightarrow\ du = -2rdr \\
r = 0 \rightarrow u = 2 \\
r = 1 \rightarrow u = 1
\end{cases}
$$

$$
\begin{aligned}
\text{Volume}(E) &= -\frac{1}{2}\int_0^{2\pi}\int_2^1 \sqrt{u}\,du\,d\theta - \int_0^{2\pi}\frac{r^4}{4}\Big|_0^1 d\theta \\
&= \frac{1}{2}\int_0^{2\pi}\frac{2}{3}u^{\frac{3}{2}}\Big|_1^2 d\theta - \int_0^{2\pi}\frac{1}{4}\,d\theta \\
&= \frac{1}{3}\int_0^{2\pi}(2\sqrt{2}-1)\,d\theta - \frac{1}{4}\cdot 2\pi \\
&= (\frac{2\sqrt{2}}{3}-\frac{1}{3})2\pi - \frac{\pi}{2} \\
&= \frac{4\sqrt{2}\pi}{3} - \frac{\pi}{3} - \frac{\pi}{2} \\
&= \frac{8\sqrt{2}\pi - 7\pi}{6} = \frac{\pi(8\sqrt{2}-7)}{6}
\end{aligned}
$$

9. Find the volume of the solid in the first octant that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, and the planes $z = 0$ and $z = 4$.

In cylindrical coordinates, the region of integration E is bounded by the planes $z = 0$ and $z = 4$. The projection of the first cylinder in the first octant onto the xy-plane is a quarter circle in the first quadrant centered at the origin with radius $r = 1$, and the projection of the second cylinder in the first octant onto the xy-plane is a quarter circle in the first quadrant centered at the origin with radius $r = 2$. Hence, E is given by

$$E = \{(r,\theta,z) : 0 \le \theta \le \frac{\pi}{2}, 1 \le r \le 2, 0 \le z \le 4\}$$

$$
\begin{aligned}
\text{Volume}(E) &= \iiint_E dV \\
&= \int_0^{\frac{\pi}{2}} \int_1^2 \int_0^4 r\,dz\,dr\,d\theta \\
&= \int_0^{\frac{\pi}{2}} \int_1^2 rz\Big|_0^4 \,dr\,d\theta \\
&= \int_0^{\frac{\pi}{2}} \int_1^2 4r\,dr\,d\theta \\
&= \int_0^{\frac{\pi}{2}} 2r^2\Big|_1^2 \,d\theta \\
&= \int_0^{\frac{\pi}{2}} (8-2)\,d\theta \\
&= \int_0^{\frac{\pi}{2}} 6\,d\theta \\
&= 3\pi.
\end{aligned}
$$

10. Evaluate the iterated integral

$$\int_{-3}^{3} \int_0^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} \sqrt{x^2+y^2}\,dz\,dy\,dx$$

by changing to cylindrical coordinates.

Letting $x = r\cos(\theta)$ and $y = r\sin(\theta)$, and $x^2 + y^2 = r^2$, we obtain

$$
\begin{cases}
z = 9 - x^2 - y^2 \implies z = 9 - r^2\cos^2(\theta) - r^2\sin^2(\theta) = 9 - r^2(\cos^2(\theta) + \sin^2(\theta)) = 9 - r^2. \\
0 \le z \le 9 - x^2 - y^2 \implies 0 \le z \le 9 - r^2
\end{cases}
$$

Hence, the region of integration is the region above the plane $z = 0$, and below the paraboloid $z = 9 - x^2 - y^2$, or $z = 9 - r^2$. Also, given $-3 \le x \le 3$ and $0 \le y \le \sqrt{9 - x^2}$, the projection of the solid region onto the xy-plane is the upper half of the disk in the xy-plane, centered at the origin, with radius 3.

$$E = \{(r, \theta, z) : 0 \le \theta \le \pi, 0 \le r \le 3, 0 \le z \le 9 - r^2\}$$

The function $f(x,y,z)=\sqrt{x^2+y^2}$ becomes $f(r,\theta,z)=r$. Therefore, the given integral becomes

$$\begin{aligned}\int_0^{\pi}\int_0^3\int_0^{9-r^2}\sqrt{r^2\cos^2(\theta)+r^2\sin^2(\theta)}r\,dz\,dr\,d\theta &= \int_0^{\pi}\int_0^3\int_0^{9-r^2} r^2\,dz\,dr\,d\theta\\ &= \int_0^{\pi}\int_0^3 r^2\, z\Big|_0^{9-r^2}\,dr\,d\theta\\ &= \int_0^{\pi}\int_0^3 r^2(9-r^2-0)\,dr\,d\theta\\ &= \int_0^{\pi}d\theta\int_0^3(9r^2-r^4)\,dr\\ &= \pi(3r^3-\frac{r^5}{5})\Big|_0^3\\ &= \pi\cdot\frac{162}{5}\end{aligned}$$

11. Evaluate the triple integral $\iiint_E (x^2+y^2)\,dV$ where E is the solid region between the spheres $x^2+y^2+z^2=4$ and $x^2+y^2+z^2=9$.

We first express the surface $x^2+y^2+z^2=4$ in spherical coordinates with $\rho=2$, and the surface $x^2+y^2+z^2=9$ in spherical coordinates with $\rho=3$. In spherical coordinates, the function $f(x,y,z)=x^2+y^2$ becomes

$$\begin{aligned}f(\rho,\theta,\phi)&=\rho^2\sin^2\phi\cos^2\theta+\rho^2\sin^2\phi\sin^2\theta\\ &=\rho^2\sin^2\phi\end{aligned}$$

We can describe E in spherical coordinates as follows.

$$E=\{(\rho,\theta,\phi):2\leq\rho\leq 3, 0\leq\theta\leq 2\pi, 0\leq\phi\leq\pi\}$$

and the integral becomes

$$\begin{aligned}\iiint_E(x^2+y^2)dV=\int_0^{2\pi}\int_0^{\pi}\int_2^3\rho^2\sin^2\phi\rho^2\sin\phi d\rho d\phi d\theta &= \left(\int_0^{2\pi}1d\theta\right)\left(\int_0^{\pi}\sin^3\phi d\phi\right)\left(\frac{\rho^5}{5}\Big|_2^3\right)\\ &= (2\pi)\frac{3^5-2^5}{5}\left(\int_0^{\pi}(1-\cos^2\phi)\sin\phi d\phi\right)=\frac{1688\pi}{15}\end{aligned}$$

12. Find the volume of the solid that lies within the sphere $x^2+y^2+z^2=4$, above the xy-plane, and below the cone $z=\sqrt{x^2+y^2}$.

We first express the surface $x^2+y^2+z^2=4$ in spherical coordinates with $\rho=2$. The equation of the cone $z=\sqrt{x^2+y^2}$ in spherical coordinates is $\phi=\pi/4$. We can describe E in spherical coordinates as follows.

$$E = \left\{(\rho, \theta, \phi) : 0 \le \rho \le 2, 0 \le \theta \le 2\pi, \frac{\pi}{4} \le \phi \le \frac{\pi}{2}\right\}$$

and the volume

$$V(E) = \iiint_E dV = \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_0^2 \rho^2 \sin\phi d\rho d\phi d\theta = \left(\int_0^{2\pi} d\theta)(\int_{\frac{\pi}{4}}^{\pi/2} \sin\phi \; d\phi)(\frac{\rho^3}{3}\Big|_0^2 \right)$$

$$= (2\pi)\frac{8}{3}(-\cos\phi|_{\frac{\pi}{4}}^{\frac{\pi}{2}}) = \frac{16\pi}{3}\left(-0 + \frac{\sqrt{2}}{2}\right) = \frac{8\sqrt{2}\pi}{3}$$

13. Find the volume of the part of the ball $\rho \le a$ that lies between the cones $\phi = \frac{\pi}{6}$ and $\phi = \frac{\pi}{3}$.

The solid region is given by

$$E = \left\{(\rho, \theta, \phi) \mid 0 \le \rho \le a, 0 \le \theta \le 2\pi, \frac{\pi}{6} \le \phi \le \frac{\pi}{3}\right\}$$

and its volume is

$$V = \iiint_E dV = \int_{\pi/6}^{\pi/3} \int_0^{2\pi} \int_0^a \rho^2 \sin\phi d\rho d\theta d\phi = \int_{\pi/6}^{\pi/3} \sin\phi d\phi \int_0^{2\pi} d\theta \int_0^a \rho^2 d\rho$$

$$= [-\cos\phi]_{\pi/6}^{\pi/3}[\theta]_0^{2\pi}\left[\frac{1}{3}\rho^3\right]_0^a = \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\right)(2\pi)\left(\frac{1}{3}a^3\right) = \frac{\sqrt{3}-1}{3}\pi a^3$$

14. Evaluate the following integrals by switching to spherical coordinates.

(a) $\displaystyle\int_{-a}^{a} \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} \int_{-\sqrt{a^2-x^2-y^2}}^{\sqrt{a^2-x^2-y^2}} (x^2 z + y^2 z + z^3)\, dz\, dx\, dy.$

From the bounds, we can see that:

$$\begin{aligned} -\sqrt{a^2 - x^2 - y^2} &\le z \le \sqrt{a^2 - x^2 - y^2} \\ -\sqrt{a^2 - y^2} &\le x \le \sqrt{a^2 - y^2} \\ -a &\le y \le a \end{aligned}$$

describes a region E that is the entire ball of radius a.
The region of integration is the solid sphere $x^2 + y^2 + z^2 \le a^2$ so $0 \le \theta \le 2\pi, 0 \le \phi \le \pi$, and $0 \le \rho \le a$. Also $f(x, y, z) = x^2 z + y^2 z + z^3 = \left(x^2 + y^2 + z^2\right) z = \rho^2 z = \rho^3 \cos\phi$. In spherical coordinates, the integral becomes

$$\int_0^{\pi} \int_0^{2\pi} \int_0^a \left(\rho^3 \cos\phi\right) \rho^2 \sin\phi d\rho d\theta d\phi = \int_0^{\pi} \sin\phi\cos\phi d\phi \int_0^{2\pi} d\theta \int_0^a \rho^5 d\rho = \left[\frac{1}{2}\sin^2\phi\right]_0^{\pi} [\theta]_0^{2\pi} \left[\frac{1}{6}\rho^6\right]_0^a = 0$$

(b) $\displaystyle\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} (x^2 + y^2 + z^2)^{3/2}\, dz\, dy\, dx.$

The integration bounds of the variables give us the region E:

$$\begin{aligned} 2-\sqrt{4-x^2-y^2} &\leq z \leq 2+\sqrt{4-x^2-y^2} \\ -\sqrt{4-x^2} &\leq y \leq \sqrt{4-x^2} \\ -2 &\leq x \leq 2 \end{aligned}$$

By using the bounds for z, we find the equation of a sphere of radius 2 with center $(0,0,2)$.

$$\begin{aligned} z =& 2+\sqrt{4-x^2-y^2} \\ (z-2)^2 &= 4-x^2-y^2 \\ x^2+y^2+(z-2)^2 &= 4 \end{aligned}$$

Because this sphere is above the xy-plane, we have that $0 \leq \phi \leq \pi/2$. Then, we express the equation of the sphere $x^2+y^2+z^2-4z+4=4$ using spherical coordinates:

$$\Rightarrow \quad \begin{aligned} \rho^2 - 4\rho\cos\phi &= 0 \\ \rho &= 4\cos\phi \end{aligned}$$

Thus, $0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \frac{\pi}{2}$, and $0 \leq \rho \leq 4\cos\phi$. Also, $f(x,y,z) = \left(x^2+y^2+z^2\right)^{3/2} = \left(\rho^2\right)^{3/2} = \rho^3$, so the integral becomes

$$\begin{aligned} \int_0^{\pi/2}\int_0^{2\pi}\int_0^{4\cos\phi} \left(\rho^3\right)\rho^2\sin\phi\, d\rho d\theta d\phi &= \int_0^{\pi/2}\int_0^{2\pi} \sin\phi\left[\frac{1}{6}\rho^6\right]_{\rho=0}^{\rho=4\cos\phi} d\theta d\phi \\ &= \frac{1}{6}\int_0^{\pi/2}\int_0^{2\pi} \sin\phi\left(4096\cos^6\phi\right) d\theta d\phi \\ &= \frac{1}{6}(4096)\int_0^{\pi/2}\cos^6\phi\sin\phi\, d\phi \int_0^{2\pi} d\theta \\ &= \frac{2048}{3}\left[-\frac{1}{7}\cos^7\phi\right]_0^{\pi/2} [\theta]_0^{2\pi} \\ &= \frac{2048}{3}\left(\frac{1}{7}\right)(2\pi) = \frac{4096\pi}{21} \end{aligned}$$

15. A solid lies inside the sphere $x^2+y^2+z^2=4z$ and outside the cone $z=\sqrt{x^2+y^2}$. Write a description of the solid using inequalities involving spherical coordinates.

The sphere $x^2+y^2+z^2=4z \Leftrightarrow x^2+y^2+z^2-4z+4=4 \Leftrightarrow x^2+y^2+(z-2)^2=2^2$ is a sphere with radius 2 centered at $(0,0,2)$.
In spherical coordinates, we have $\rho^2 = 4\rho\cos\phi \Leftrightarrow \rho^2-4\rho\cos\phi=0 \Leftrightarrow \rho=0$ or $\rho=4\cos\phi$, so "inside the sphere" is described by $0 \leq \rho \leq 4\cos\phi$.
The cone $z=\sqrt{x^2+y^2}$ is described by $\phi=\frac{\pi}{4}$, so "outside the cone" is described by $\frac{\pi}{4} \leq \phi \leq \frac{\pi}{2}$.
We can describe the solid region E in spherical coordinates as follows.

$$E = \left\{(\rho,\theta,\phi) : 0 \leq \theta \leq 2\pi, \frac{\pi}{4} \leq \phi \leq \frac{\pi}{2}, 0 \leq \rho \leq 4\cos\phi\right\}$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Cylindrical Coordinates. In the cylindrical coordinate system, a point P in space is represented by an ordered triple (r, θ, z), where (r, θ) gives the polar coordinates of the projection of P onto the xy-plane, and z is the same z-coordinate as in rectangular coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

If a solid E can be described in cylindrical coordinates by

$$E = \{(r, \theta, z) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq r \leq h_2(\theta),\ u_1(r, \theta) \leq z \leq u_2(r, \theta)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta.$$

Cylindrical coordinates can also be defined using polar coordinates in the yz-plane or the xz-plane instead of the xy-plane, and in each case the Jacobian factor is still r.

Spherical Coordinates. In the spherical coordinate system, a point P in space is represented by an ordered triple (ρ, θ, ϕ), where ρ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment from the origin to P:

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

If a solid E can be described in spherical coordinates by

$$E = \{(\rho, \theta, \phi) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq \phi \leq h_2(\theta),\ u_1(\theta, \phi) \leq \rho \leq u_2(\theta, \phi)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(\theta,\phi)}^{u_2(\theta,\phi)} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\, \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta.$$

Suggested Textbook Problems
Section 15.6: 1-47, 57, 58
Section 15.7: 1-32
Section 15.8: 1-32,37-40,43-46